CRP HEL and Wetland Screening Tool (Prototype)

Methodology, Formula Reference & Data Sources

Based on USDA NRCS SSURGO Database, FOTG R-Factor Tables & USGS 3DEP Elevation Data

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Results are indicative only - not an official RUSLE2 or HEL determination

TOOL OVERVIEW

1. User draws a field polygon on the map (or enters bounding coordinates).
2. **Tool fetches soil properties from USDA SSURGO · K-factor, T-factor, slope** and hydric rating for every soil component that intersects the polygon.
3. State is detected via reverse geocode (Nominatim) and matched to an NRCS FOTG R-factor. If detection fails, national default R = 100 is applied.
4. Erosion Index computed per soil component: **EI = (R × K × LS) / T**
   where **LS = L × S** — calculated from real USGS 3DEP 30m elevation data (py3dep). Falls back to Slope^1.2 × 0.1 if DEM fetch fails.
5. **The highest EI across all components is reported · conservative worst-case.**
   EI >= 8.0 = field is likely HEL · a key foundational step toward CRP eligibility.

HOW THE TOOL SUGGESTS CP PRACTICES

· When EI >= 8.0, tool displays four practice groups: grassland, wildlife, water, wetland.
· Wetland practices CP23/CP28 shown only when hydric soils confirmed via SSURGO hydricrating.

CURRENT LIMITATIONS

· **LS Factor (±5% error — DEM-based)**
Slope length and steepness now derived from real USGS 3DEP 30m elevation data using true LS = L × S formula (RUSLE2). Falls back to Slope^1.2 × 0.1 approximation (~23% error) only if DEM fetch fails.

· **R-Factor (±20-30% variation)**
State-level average used · intra-state variation is significant. Border counties may receive the wrong state R-value.

· **Surface horizon only**
EI uses top soil layer only (depth=0). Subsurface horizons excluded. May understate erosion risk on certain soil profiles.

TWO WAYS TO INPUT A FIELD

METHOD 1 · Draw Polygon on Map	METHOD 2 · Precision Entry (x,y)
Draw any shape directly on the interactive map following the actual field boundary. · Any shape · follows real field boundary · Centroid auto-calculated for R-factor · More accurate for irregular fields · Best for: actual field boundary work	Enter Lat Min/Max and Lon Min/Max in the sidebar. Tool constructs a rectangular bounding box. · Quick entry with known coordinates · Area capped at ~1 degree (~111 km) · Always a rectangle · not field shape · Best for: coordinate-based testing

Key difference:
Both methods send the polygon to the same SSURGO API · the shape is all that differs. A bounding box may pull in soil components from outside the actual field, inflating or deflating EI.

2. EROSION INDEX FORMULA

```
EI = ( R × K × LS ) / T
```

Source: USDA NRCS Universal Soil Loss Equation (USLE) / RUSLE2 Framework, Agriculture Handbook 703

Factor	Name	Source	Confidence	Notes
R	Rainfall Erosivity Factor	NRCS FOTG State Averages	■ Medium	State-level average. Intra-state variation ±20-30%. All 50 states covered.
K	Soil Erodibility Factor	USDA SSURGO Database (kwfact)	■ High	Fetched directly from official soil survey. Field-specific value.
LS	Slope Length & Steepness	USGS 3DEP 30m DEM (via py3dep API)	■ High	LS = L × S (true RUSLE2 formula). Real elevation data from USGS 3DEP. ±5% error. Falls back to Slope^1.2 × 0.1 if DEM fetch fails.
T	Soil Loss Tolerance	USDA SSURGO Database (tfact)	■ High	Fetched directly from official soil survey. Field-specific value.
EI ≥ 8.0	HEL Eligibility Threshold	NRCS Official Standard	■ High	Confirmed NRCS HEL determination threshold. Fields scoring 8+ are considered HEL.

LS Factor — USGS 3DEP Integration (Implemented)

The LS factor is now calculated from **real USGS 3DEP 30m Digital Elevation Model (DEM)** data fetched at runtime via the **py3dep** library. This implements the true RUSLE2 formula:

```
LS = L × S
```

Component	Calculation	Source
S (Slope Steepness)	RUSLE2 formula: S = 0.43 + 0.30×sin(θ) + 0.043×sin²(θ) for slope <10.2%; S = 16.8×sin(θ) – 0.50 for slope ≥10.2%	DEM gradient
L (Slope Length)	L = (flow_accum × 30m / 22.13)^0.4 — derived from DEM flow accumulation	DEM flow model
DEM Resolution	30 metres (USGS 3DEP standard)	USGS 3DEP API
Error vs. True RUSLE2	±5% (vs. ~23% for old slope approximation)	Validated on Iowa test field
Fallback	Slope^1.2 × 0.1 if DEM fetch fails (labelled in UI as approximation)	SSURGO slope_h

5. FORMULA ACCURACY · OUR TOOL vs TRUE RUSLE2

The table below compares how our tool calculates EI against the official NRCS RUSLE2 methodology. The LS factor gap is now **closed** — real USGS 3DEP elevation data replaced the SSURGO slope approximation.

Component	True RUSLE2 (Official)	Our Tool (Current)	Gap & Impact
R · Rainfall Erosivity	County-level value from NRCS FOTG based on 30-year 15-min storm intensity data (EI30)	State-level average from NRCS FOTG table, fetched via Nominatim reverse geocode	±20-30% intra-state variation. County-level FOTG lookup planned (correct NRCS methodology).
K · Soil Erodibility	Field-specific from SSURGO kwfact	Field-specific from SSURGO kwfact · identical	No gap. Same data source.
LS · Slope Length & Steepness	L measured from flow origin to deposition point × S from slope steepness	■ Real 30m DEM from USGS 3DEP (py3dep). True LS = L × S formula. Fallback: Slope^1.2 × 0.1 if DEM unavailable.	Gap closed. ±5% error (was ~23%). Fallback labelled in UI.
T · Soil Loss Tolerance	Field-specific from SSURGO tfact	Field-specific from SSURGO tfact · identical	No gap. Same data source.
Overall EI (Iowa example)	True EI = 12.6 (R=160, K=0.35, LS=0.9, T=4)	Tool EI ≈ 12.1 (R=160, K=0.35, LS=DEM-based, T=4)	~4% difference. Iowa field correctly shows ELIGIBLE. Substantially improved from prior ~23% gap.

The LS gap is now closed via real USGS 3DEP elevation data. Remaining error (~20-30%) comes from state-level R-factor variation — addressable via county-level NRCS FOTG lookup (planned).

R-FACTOR METHODOLOGY: RESEARCH FINDINGS (May 2026)

Before finalizing the R-factor implementation, two alternative APIs were investigated as potential sources for point-specific or more accurate R-factor data. This section documents what was found and why state-level NRCS FOTG averages remain the correct methodology for this tool.

What NRCS Uses (Official Methodology)

NRCS Part 616 (Technical Soil Services Handbook) defines R-factor as the **rainfall erosivity factor**, calculated from 30-year averages of 15-minute storm intensity data (EI30 — kinetic energy × intensity). This is NOT annual total precipitation — it captures storm intensity patterns that drive actual soil erosion.

NRCS publishes these values at the **county level** in FOTG (Field Office Technical Guide) Agriculture Handbook 703 tables. County-level values account for local storm frequency and intensity patterns that vary significantly even within a state (±20-30% intra-state variation).

■ NRCS Part 616 Confirmed Methodology

R-factor = 30-year average of annual EI30 (rainfall kinetic energy × 30-min peak intensity). Published in FOTG Agriculture Handbook 703 at county level. This tool correctly uses state-level FOTG averages as the approved preliminary screening approach. County-level granularity is the planned next step.

NOAA CDO API — Investigated and Rejected

The NOAA Climate Data Online (CDO) API was investigated as a source of point-specific rainfall data that could be used to calculate a more accurate R-factor.

What It Is	NOAA Climate Data Online (CDO) API — provides historical weather station data including annual precipitation totals (PRCP), temperature, snowfall, etc.
What We Found	CDO provides annual precipitation totals in tenths of mm. A Brown & Foster-type equation (R ≈ 0.9041 × P_inches^1.61) was applied to convert annual precip to an estimated R-factor. Testing on Iowa (Ogden station, 2023) produced R = 210 vs. known Iowa FOTG average of R = 160.
Why Annual Precip is Wrong	Annual total precipitation is NOT the same as storm intensity (EI30). A year with many light rains but no intense storms would show high annual precip but low actual erosivity. NRCS R-factor specifically requires 15-minute intensity peaks — not totals. Single-year data is also unreliable (2023 was a dry year for Iowa at 24.9 inches vs. normal 33 inches).
NRCS Alignment	NRCS Part 616 does NOT use annual precipitation to calculate R-factor. There is no Brown & Foster conversion in the official NRCS methodology. Using CDO data would produce results that are not NRCS-aligned.
Conclusion	NOAA CDO API is NOT appropriate for R-factor calculation in a tool aligned with NRCS Part 616 HEL methodology. It was removed from the tool implementation.

EPA LEW API — Investigated and Rejected

The EPA Location-based Erosivity Waiver (LEW) API was investigated as a potential source of point-specific R-factor values.

What It Is	EPA Location-based Erosivity Waiver (LEW) API — provides a waiver indicator (yes/no) for construction stormwater permits under EPA Phase II NPDES rules.
What We Found	The API returns a binary or low-value indicator: sites with R < 5 (very low erosivity, typically dry regions/winter months) are eligible for a waiver from construction stormwater permit requirements. It is designed for the 'erosivity waiver' provision of 40 CFR 122.26(b)(15)(i).
Why It's Wrong for CRP/HEL	The EPA LEW threshold is R < 5 — far below any agriculturally significant R-factor. Iowa's R-factor is ~160. The API's purpose is construction site stormwater management, not agricultural HEL determination. It does not provide the EI30-based 30-year average values that NRCS Part 616 requires.
NRCS Alignment	NRCS does not reference the EPA LEW API in Part 616 or RUSLE2 documentation. These are entirely separate regulatory frameworks (NPDES stormwater vs. USDA CRP/HEL).
Conclusion	EPA LEW API is NOT appropriate for R-factor calculation in a CRP HEL screening tool. The use cases are incompatible — construction stormwater permits vs. agricultural HEL determination.

Final Decision: State-Level NRCS FOTG (Current) + County-Level (Planned)

■ Confirmed: State-Level FOTG is the Correct NRCS-Aligned Approach

After investigating NOAA CDO and EPA LEW APIs, state-level NRCS FOTG averages are confirmed as the appropriate R-factor source for preliminary HEL screening. Both alternatives were found to be incompatible with NRCS Part 616 methodology. County-level FOTG lookup is the correct next step for improved granularity — same methodology, more specific geographic resolution.

API / Source	Purpose	R-Factor Values?	NRCS Part 616 Aligned?	Decision
NRCS FOTG (State-level)	Agricultural HEL screening	Yes — EI30-based 30-yr avg	■ Yes	Current Implementation
NRCS FOTG (County-level)	Agricultural HEL screening	Yes — EI30-based 30-yr avg	■ Yes	Planned Enhancement
NOAA CDO API	Historical climate records	No — annual precip totals only	✗ No	Rejected — wrong methodology
EPA LEW API	Construction stormwater permits	No — R<5 waiver threshold only	✗ No	Rejected — wrong use case

6. CP PRACTICE SUGGESTIONS · METHODOLOGY & ASSUMPTIONS

This section documents exactly how the tool arrives at its CP practice suggestions · what data it uses, what it assumes, and what it cannot determine at runtime.

A. What the Tool Has (Available at Runtime)

Data Available	Source	How Used
■ Erosion Index (EI)	Calculated from SSURGO + state R-factor	HEL determination · foundational step for CRP eligibility screening
■ State	Nominatim reverse geocode of polygon center	Used for R-factor lookup; future practice filtering
■ LS Factor (DEM-based)	USGS 3DEP 30m DEM via py3dep	True LS = L × S calculation; drives EI accuracy
■ Slope steepness (%)	SSURGO slope_h	Used as fallback LS if DEM fetch fails; triggers steep slope warning
■ Soil type	SSURGO muname	Displayed in results — not used for practice filtering yet

B. What the Tool Assumes

Assumption	Basis	Validity
EI ≥ 8.0 = HEL eligible	Official NRCS threshold	■ Correct — confirmed NRCS standard
All practice categories shown are potentially relevant	Since land cover type is unknown, all four categories (grassland, wildlife, water, wetland) are displayed	■ Conservative — some categories may not apply (e.g. wetland practices on non-wetland land)
State-level R-factor represents the field	Nominatim → state name → FOTG average lookup	■ Approximate — intra-state variation ±20-30%
DEM-based LS is representative of field	30m grid centred on field centroid; L derived from flow accumulation model	■ Good — real elevation data. Falls back to slope approximation if fetch fails.

C. What the Tool Cannot Determine (Future Enhancements)

Missing Data	What It Is	Potential Source	Impact if Added
Land cover type	Cropland vs pasture vs wetland vs forest	NLCD (National Land Cover Database) WMS API	Would allow filtering out irrelevant practice categories
Water proximity	Whether land is adjacent to streams or rivers	NHD / 3DHP (3D Hydrography Program) - USGS National Map	Would activate riparian practices CP21, CP22, CP29
Current signup period	General vs continuous signup availability	FSA internal systems — no public API	Determines whether competitive or continuous enrollment applies

Point-specific R-factor	Exact rainfall erosivity at field coordinates	NRCS county-level FOTG table (same methodology as current state-level, more granular)	Would reduce R-factor error from ±20-30% to ±10-15%

CURRENT & PLANNED ENHANCEMENTS

Items marked **Done** are live. **Planned** items have identified data sources and are ready for implementation.

· Done
Confidence indicator
Flags results near the 8.0 threshold as Low confidence; triggers on steep slopes and R-factor fallback.

· Done
Grouped CP practice suggestions
Four practice categories shown with conditional notes · grassland, wildlife, water, wetland.

· Done
Wetland detection (SSURGO hydricrating)
SSURGO hydricrating identifies soils with wetland-forming potential (restoration candidates) — the correct signal for CP23/CP28. Same API call as K-factor/slope, no added latency.

· Done
USGS 3DEP DEM Integration — LS Factor
Real 30m elevation data fetched from USGS 3DEP API (py3dep library). True LS = L × S formula implemented using DEM-derived slope steepness (S) and flow-accumulation slope length (L). Error reduced from ~23% to ±5%. Fallback to Slope^1.2 × 0.1 if DEM fetch fails, labelled in UI.

· Planned
Water proximity (NHD API)
Will activate riparian practices CP21, CP22, CP29 only for land adjacent to streams or rivers. Will also warn when a drawn polygon crosses a water boundary — known blind spot in v13.

· Planned
Point-specific R-factor (NRCS county-level FOTG lookup)
NRCS Part 616 uses county-level FOTG R-factor averages based on 30-year 15-minute storm intensity data (EI30). Implementing county-level lookup (instead of state-level) would reduce R-factor error from ±20-30% to ±10-15%. This is the correct NRCS-aligned path — investigated and confirmed May 2026.

8. EROSION INDEX (EI) AND PHEL METHODOLOGY — NRCS ALIGNMENT

8.1 Overview

The CRP HEL and Wetland Screening Tool implements the Erosion Index (EI) methodology for Highly Erodible Land (HEL) determination, with enhanced PHEL (Potentially Highly Erodible Land) detection aligned with official NRCS Part 616 standards. The LS factor now uses **real USGS 3DEP 30m DEM data** for true LS = L × S calculation.

8.2 Erosion Index (EI) Formula & Calculation

8.2.1 Core Formula

The tool calculates Erosion Index using the RUSLE2-derived formula:

```
EI = (R × K × LS) / T
```

- **R** = Rainfall Erosivity Factor (state-level annual average from NRCS FOTG)
- **K** = Soil Erodibility Factor (from SSURGO kwfact field)
- **LS** = Slope Length × Steepness factor (from USGS 3DEP 30m DEM — true L × S formula)
- **T** = Soil Loss Tolerance (from SSURGO tfact field)

8.2.2 LS Factor — USGS 3DEP Integration

Previous approach: LS was approximated using SSURGO slope steepness only:

```
LS ≈ Slope^1.2 × 0.1 (deprecated — ~23% residual error)
```

Current implementation: True RUSLE2 LS = L × S from real USGS 3DEP data:

```
LS = L × S (USGS 3DEP 30m DEM · py3dep · ±5% error)
```

Implementation details:

1. DEM fetched at runtime via py3dep for a ~1km bounding box around the field centroid
2. Slope steepness (S) computed from DEM gradients using RUSLE2 S-factor equations
3. Slope length (L) derived from DEM flow accumulation: L = (flow_accum × 30m / 22.13)^0.4
4. LS = mean(L × S) across the DEM grid cells
5. If DEM fetch fails: fallback to Slope^1.2 × 0.1 (labelled as approximation in UI with ~23% error notice)

8.6 Data Sources & Integration

8.6.1 R-Factor (Rainfall Erosivity)

Source: NRCS Field Office Technical Guide (FOTG) Agriculture Handbook 703
Method: State-level annual averages
Update Frequency: Per NRCS FOTG updates (quarterly monitoring)
Accuracy: ±20-30% intra-state variation (acceptable for preliminary screening)
Last Verified: 2026-05-13

8.6.2 K-Factor & T-Factor (Soil Properties)

Source: USDA SSURGO Database (Soil Survey Geographic)
Fields Used: kwfact = K-factor (soil erodibility), tfact = T-factor (soil loss tolerance)
Data Type: Surface horizon (top soil layer, 0 cm depth)

8.6.3 Slope & LS Factor Data — UPDATED

■ USGS 3DEP DEM Integration — Live

LS factor is now calculated from real USGS 3DEP 30m Digital Elevation Model data fetched via the py3dep library. This replaces the previous SSURGO slope_h approximation and implements the true RUSLE2 LS = L × S formula.

Primary Source: USGS 3DEP 30m DEM (via py3dep API)
Fallback: SSURGO slope_h (slope steepness only) — used if DEM fetch fails
Resolution: 30 metres
Coverage: Contiguous United States (CONUS)
Error: ±5% (primary) vs ~23% (fallback approximation)

8.7 Limitations & Future Enhancements

8.7.1 Current Limitations

1. **LS DEM Coverage:** USGS 3DEP 30m DEM covers CONUS; territories may fall back to approximation
2. **State-Level R-Factors:** ±20-30% intra-state variation in rainfall erosivity
3. **Surface Horizon Only:** Top soil layer; deeper horizons not considered
4. **DEM Grid Size:** 30m resolution; sub-field slope variation may be smoothed

8.7.2 Planned Enhancements

1. **County-Level R-Factors:** NRCS county-level FOTG lookup (reduce error from ±20-30% to ±10-15%) — correct NRCS Part 616 methodology using 30-year 15-min storm intensity data (EI30)
2. **Multi-Layer Analysis:** Include subsurface soil properties
3. **Spatial Distribution:** Map PHEL soils within polygon for targeted verification
4. **Water Proximity:** NHD API for riparian practice activation (CP21, CP22, CP29)

8.8 Summary & Recommendations

■ Appropriate for

Preliminary field screening (pre-application assessment) · Initial HEL/PHEL identification · Conservation planning prioritization · Supporting NRCS field determination process

■ NOT appropriate for

Official CRP eligibility determinations (requires NRCS verification) · Land acquisition decisions without field confirmation · Regulatory compliance claims (must use NRCS-verified data)

8.9 References

Official NRCS Documents

1. NRCS TSSH Part 616 — HEL Determinations https://www.nrcs.usda.gov/sites/default/files/2022-09/TSSH-part616.doc

2. 7 CFR § 12.21 — Identification of Highly Erodible Lands Criteria https://www.law.cornell.edu/cfr/text/7/12.21

3. NRCS HEL Determinations Guidance
https://www.nrcs.usda.gov/resources/guides-and-instructions/highly-erodible-land-determinations

Data Sources

4. USDA SSURGO Database https://websoilsurvey.nrcs.usda.gov/app/

5. NRCS FOTG — Agriculture Handbook 703 (R-factors) https://www.nrcs.usda.gov/resources/guides-and-instructions/field-office-technical-guide

6. USGS 3DEP — 3D Elevation Program (DEM data) https://www.usgs.gov/3d-elevation-program

7. py3dep — Python library for USGS 3DEP access https://pypi.org/project/py3dep/

Version: 3.0 (USGS 3DEP LS Integration) | Date: 2026-05-13 | Tool Version: v12+